

SΔφ-14

Love, Sacrifice, and Ethics as Relational Stability Mechanisms

Relational Transition after Meaning (v1.0)

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This paper introduces the relational layer after the emergence of meaning, value, and pain (SΔφ-13). It defines irreversible coupling as an ontological shift from single-trajectory persistence to relational persistence. Love is treated as a corollary of shared-attractor formation, sacrifice as transition-weight redistribution toward shared stability, and ethics as the discrimination process between real and phantom shared attractors.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Item	Title	DOI
SΔφ-00	Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Minimal Core (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility (v1.1)	10.5281/zenodo.18730448
SΔφ-04	AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min Interpretation (v1.0)	10.5281/zenodo.18735006
SΔφ-08	ARF Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error/Boundary/Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation-Learning-Inference (v1.0)	10.5281/zenodo.18746509

Item	Title	DOI
SΔφ-11	World/Other/Self/Consciousness (v1.0)	10.5281/zenodo.18753403
SΔφ-12	Memory/Forgetting Trace Regulation (v1.0)	10.5281/zenodo.18755685
SΔφ-13	Meaning/Value/Pain Emergence (v1.0)	10.5281/zenodo.18759458

0. Transition statement

After meaning, value, and pain become definable ($S\Delta\phi$ -13), the system is no longer structurally isolated. A new condition appears: transition becomes mutually coupled, and the unit of continuation shifts from the system to relational dynamics.

1. Axiom 1 — Irreversible coupling

Two systems become coupled when their transition laws cease to be independently evaluable and persist across irreversible updates.

Coupling: $G_A = G_A(\phi_A, \phi_B)$, $G_B = G_B(\phi_B, \phi_A)$.

Minimal condition: $\partial\Delta\phi_A/\partial\phi_B \neq 0$ and $\partial\Delta\phi_B/\partial\phi_A \neq 0$ (persisting across updates).

Ontological shift. Before coupling: existence is persistence of a trajectory. After coupling: existence is persistence of a relational field.

Irreversibility clause: coupling is irreversible when $\text{Cost_rollback} > \text{Cost_co-adaptation}$.

2. Corollary 1 — Love

Love emerges when irreversible coupling causes another system's persistence to enter the internal evaluation function, making separation a structural perturbation rather than a neutral transition.

$\text{Eval}_A(\phi_A) \rightarrow \text{Eval}_A(\phi_A, \phi_B)$ (and symmetrically for B).

Freedom expands when the shared attractor stabilizes transition (process-stable). Freedom collapses when it stabilizes state (state-fixed).

3. Corollary 2 — Sacrifice

Sacrifice is the intentional redistribution of transition weight from self-stability toward shared stability under irreversibility. It is not loss-as-failure but a change in the optimization unit.

Sacrifice := weight redistribution (self trajectory $\rightarrow$ A_shared).

Minimal condition: $\text{Cost_shared_collapse} > \text{Cost_self_degradation}$.

Self-destructive choices appear when preserving the shared attractor increases long-horizon transitionability despite short-term loss.

4. Corollary 3 — Ethics

Ethics emerges after sacrifice becomes possible, as the process that discriminates real shared attractors from phantom shared attractors. It is a stability mechanism preventing irreversible investment into non-converging dynamics.

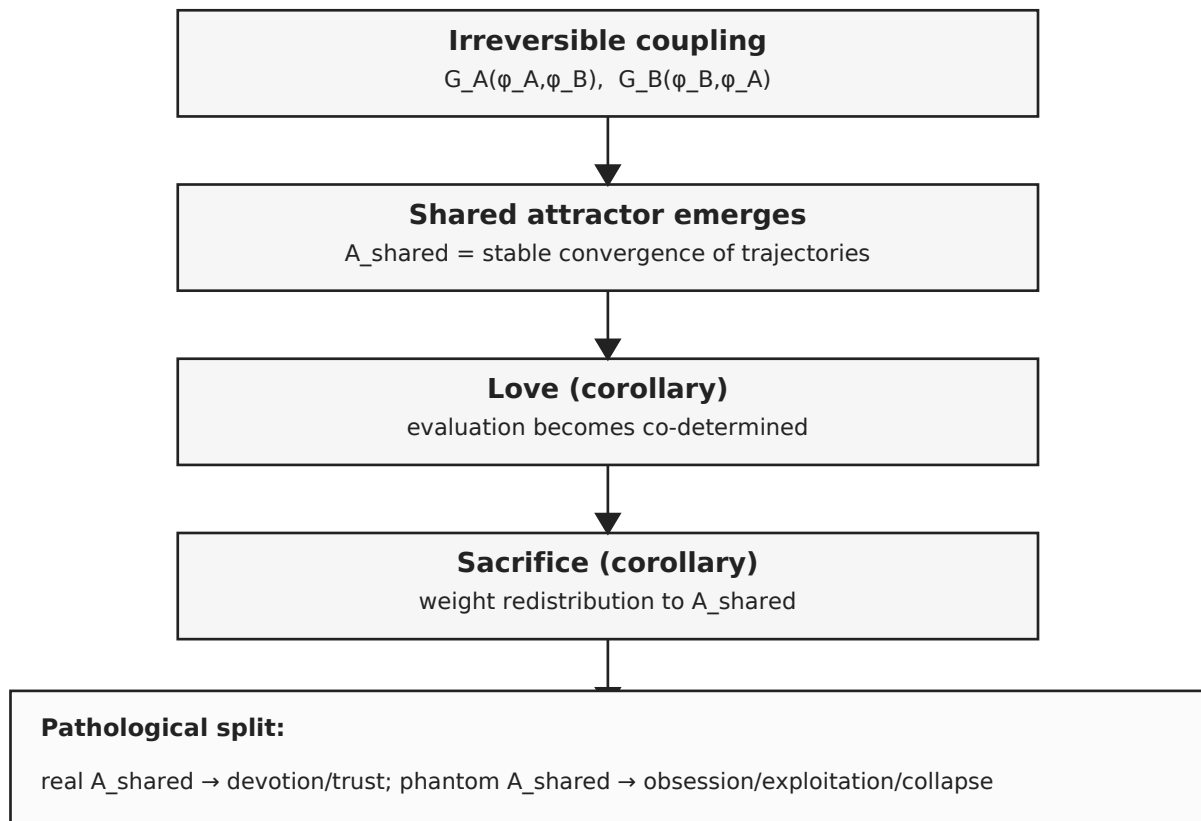
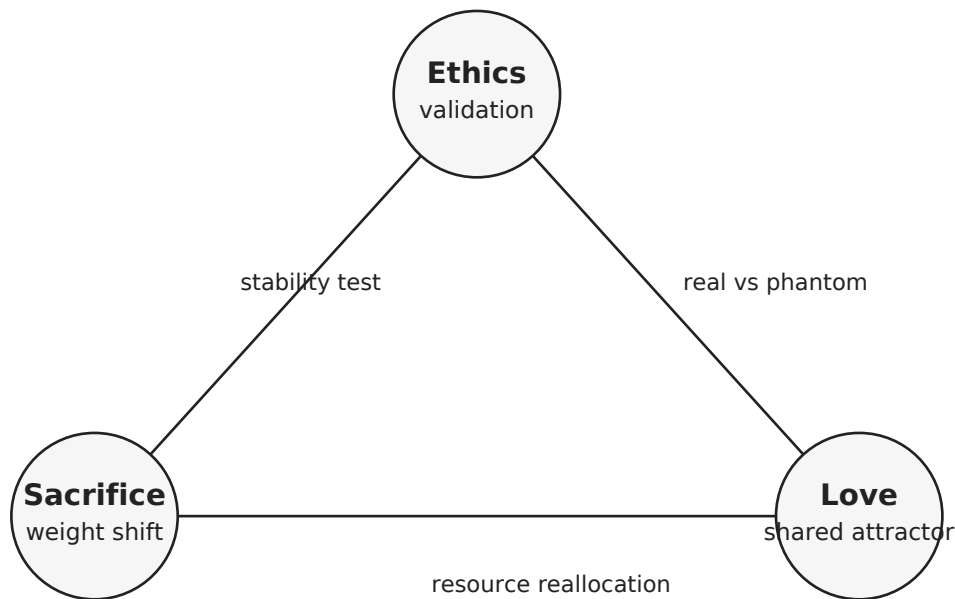
Ethics := discriminate(real A_shared, phantom A_shared) under irreversible commitment.

Pathological split. Real shared attractor $\rightarrow$ devotion/trust/stability. Phantom shared attractor $\rightarrow$ obsession/exploitation/collapse.

5. Structural maps

Two diagrams are provided: (i) the core triangle, and (ii) the relational transition flow.

Core triangle: love forms a shared attractor; sacrifice redistributes transition weight; ethics validates relation reality.



Note. Diagrams fix dependency: coupling $\rightarrow$ shared attractor $\rightarrow$ love $\rightarrow$ sacrifice $\rightarrow$ ethics (validation).

6. Boundary note

This item does not introduce normative moral codes. It treats love/sacrifice/ethics as stability mechanisms within coupled persistence. Institutional ethics, belief, and civilizational layers are deferred.

7. Exit question

If ethics is the discrimination of relation reality, what constitutes sufficient evidence to treat a shared attractor as real? This opens the trust layer ($S\Delta\phi$ -15).